

(a) Based on the regularized Bellman expectation equation, we can introduce the “regularized Bellman expectation operator”  $T_{\Omega}^{\pi}$  as

Please verify that  $T_{\Omega}^{\pi}$  is a contraction operator in  $\ell_{\infty}$  norm. (Hint: Try to extend the proof of the contraction property of the unregularized operator  $T^{\pi}$  described in Lecture 3)

$$V_{\Omega}^*(s) = \max_{\pi \in \Pi} \{Q_{\pi}^*(s) + \gamma P_{\pi}^* V_{\Omega}^*\} \quad (2)$$

1a.) Consider  $L^\infty$  norm to measure the distance between two value function.  $V$  &  $V'$   
 $\Rightarrow \|V - V'\|_\infty = \max_{s \in S} |V(s) - V'(s)|$ ,  $\rho_{\text{reg}}$ , i.e. "regularized  $\| \cdot \|_\infty$  expectation operator"  
 $T_\Omega^\gamma$  as,  $T_\Omega^\gamma(V) := R^\gamma + \Omega(\gamma) + \gamma P^\gamma V$   
 $\|T_\Omega^\gamma(V) - T_\Omega^\gamma(V')\|_\infty = \| (R^\gamma + \Omega(\gamma) + \gamma P^\gamma V) - (R^\gamma + \Omega(\gamma) + \gamma P^\gamma V') \|_\infty$   
 $= \|\gamma P^\gamma(V - V')\|_\infty$   
 $\leq \gamma \|V - V'\|_\infty$ , for  $\gamma \leq 1$   
 $\Rightarrow$  proof that  $T_\Omega^\gamma$  is a  $\gamma$ -contraction operator.

For Equation 12:  $V_{\pi}(s) = \max_{a \in \Gamma} \{ R_s^a + \gamma \sum_{s'} P_{s'}^a V_{\pi}(s') + \gamma H(\pi(s)) \}$

$R_s^a$  (expected reward): It is the expected reward when in state  $s$  do the policy  $\pi$   $\Rightarrow \sum_{s'} \pi(s') R(s, a, s')$

$\sum_{s'} P_{s'}^a V_{\pi}(s')$  (expected future value): It is the expected future value when take policy  $\pi$ , next state to  $s'$ 's expected value  $\Rightarrow \sum_{s'} R(s, a, s') \sum_{s''} P(s''|s, a) V_{\pi}(s'')$

- substitute the equations into equation 12, then we'll get:  $V_{\pi}(s) = \max_{a \in \Gamma} \left\{ \sum_{s'} \pi(s') R(s, a, s') + \gamma \sum_{s'} \sum_{s''} P(s''|s, a) V_{\pi}(s'') + \gamma H(\pi(s)) \right\}$

$\Rightarrow \max_{a \in \Gamma} \left\{ \sum_{s'} \pi(s') [R(s, a, s') + \gamma \sum_{s''} P(s''|s, a) V_{\pi}(s'')] + \gamma H(\pi(s)) \right\}$

$\Rightarrow \max_{a \in \Gamma} \left\{ \sum_{s'} \pi(s') [G_{\pi}^*(s, a) + \gamma H(\pi(s))] \right\}$  optimize  $\pi$  to get the maximum value

$\Rightarrow \max_{\pi \in \Pi} \left\{ \sum_{s'} \pi(s') [G_{\pi}^*(s, a) + \gamma H(\pi(s))] \right\}$

$$\Rightarrow \text{Take derivative to } \pi(a) = \beta_{(15,a)}^* - (\beta_{(15,a)}^* + 1) - \lambda = 0.$$

$$\begin{aligned} \Rightarrow \int_{-\infty}^{\infty} \tilde{\kappa}(s, \lambda) &= \hat{\rho}_{11}^*(s, \lambda) - 1 - \lambda \\ \Rightarrow \tilde{\kappa}(s, \lambda) &= \exp(\hat{\rho}_{11}(s, \lambda)) \cdot \exp(-1 - \lambda) \quad \therefore \tilde{\kappa}(s, \lambda) = 1, \text{ i.e. } \exp(-1 - \lambda) = \frac{1}{\int_{-\infty}^{\infty} \exp(\hat{\rho}_{11}(s, \lambda)) ds} \\ \Rightarrow \text{so } \tilde{\kappa}(s, \lambda) &= \frac{\exp(\hat{\rho}_{11}^*(s, \lambda))}{\int_{-\infty}^{\infty} \exp(\hat{\rho}_{11}^*(s, \lambda)) ds} \quad |s| \rightarrow \infty \quad \text{constant.} \end{aligned}$$

$$\begin{aligned} \text{For } V_{\lambda}^{(1)} &= \max_{\pi \in \Pi} \left[ \sum_{i \in A} \pi(i) \cdot \left\{ \beta_{\lambda}^{(1)} \cdot \lambda_{\pi(i)} - \lambda_{\pi(i)} | \pi(i) | \right\} \right] \quad \left| \pi^*(i) = \arg \max_{i \in A} \left\{ \beta_{\lambda}^{(1)} \cdot \lambda_i - \lambda_i |i| \right\} \right. \\ &\Rightarrow \text{N.B. that } \max_{i \in A} \exp \left\{ \beta_{\lambda}^{(1)} \cdot \lambda_i - \lambda_i |i| \right\} = \exp \left\{ \beta_{\lambda}^{(1)} \cdot \lambda_{\pi^*(i)} - \lambda_{\pi^*(i)} | \pi^*(i) | \right\} \quad \text{Take } \lambda_{\pi^*(i)} \text{ on both sides.} \\ &\Rightarrow \lambda \cdot \left( \frac{1}{\sum_{i \in A} \exp \left\{ \beta_{\lambda}^{(1)} \cdot \lambda_i - \lambda_i |i| \right\}} \right) \cdot \left( \beta_{\lambda}^{(1)} \cdot \lambda_{\pi^*(i)} - \lambda_{\pi^*(i)} | \pi^*(i) | \right), \text{ substitute it in } V_{\lambda}^{(1)} \text{ equation.} \\ V_{\lambda}^{(1)} &= \sum_{i \in A} \pi^*(i) \cdot \lambda_{\pi^*(i)} \\ &= \lambda \cdot \left( \frac{1}{\sum_{i \in A} \exp \left\{ \beta_{\lambda}^{(1)} \cdot \lambda_i - \lambda_i |i| \right\}} \right) \cdot \left( \sum_{i \in A} \exp \left\{ \beta_{\lambda}^{(1)} \cdot \lambda_i - \lambda_i |i| \right\} \right) \\ &= \lambda \cdot \left( \frac{1}{\sum_{i \in A} \exp \left\{ \beta_{\lambda}^{(1)} \cdot \lambda_i - \lambda_i |i| \right\}} \right) \cdot \text{an optimal solution for } V_{\lambda}^{(1)} \\ &= \lambda \cdot \left( \frac{1}{\sum_{i \in A} \exp \left\{ \beta_{\lambda}^{(1)} \cdot \lambda_i - \lambda_i |i| \right\}} \right) \cdot \left( \sum_{i \in A} \exp \left\{ \beta_{\lambda}^{(1)} \cdot \lambda_i - \lambda_i |i| \right\} \right) \end{aligned}$$

step 1. For every state  $s$ , set every  $V_0(s) = 0$ .

$V_{\text{pstate}} Q = Q_{\text{in}}(s, \omega) + R_{\text{in}} + r \sum_{i=1}^n p_i(1-s_i) [V_{\text{K}}(s_i)]$   
 $V_{\text{pstate}} V = V_{\text{K}}(1-s) = \int_{\text{K}} \sum_{i=1}^n p_i(s, \omega)$   
 $V_{\text{K}}(1) | V_{\text{K}}(1) = V_{\text{K}}(1 \text{ convergance}) \cdot V_{\text{K}}(1) + V_{\text{K}}(1) + V_{\text{K}}(1)$

→ Pseudo code:

- Initialize value function for all state.
- Initialize  $V(s) = 0$  for all state.
- Compute the optimal value function.
- repeat until convergence:

for each state  $s$ :  
 \* Update  $Q_s$  with  $V_k$ .  
 for all action  $a$ :  

$$Q(s, a) = R(s, a) + \gamma \sum_{s'} P(s' | s, a) \cdot V_k(s')$$

# Step 2. Update  $V_{k+1}$  with  $G_k$

$$V_{k+1}(s) = \int_a^b \sum_i \{a_i\} \exp(\beta_k(s, a_i))$$

# After convergence, calculate optimal  $Q$ .

for each state  $S$ :

for each action  $a$ :

$$Q^*(s, a) = R(s, a) + \gamma \sum_{s'} P(s' | s, a) \cdot V^*(s')$$

(a) Please prove the following useful property discussed in Lecture 4: For any function  $f: S \times \mathcal{A} \rightarrow \mathbb{R}$ ,

$$\mathbb{E}_{\tau \sim \mu^{\pi_\theta}} \left[ \sum_{t=0}^{\infty} \gamma^t f(s_t, a_t) \right] = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [f(s, a)] \quad (4)$$

(Hint: It might be slightly easier to go from the **RHS to LHS**. Specifically, you may first **expand the RHS of (4) into a sum of  $f(s, a)$  over  $s$  and  $a$**  and then apply the definition of  $d_\mu^{\pi_\theta}$ , which involves a sum of probability over  $\tau$ . Next, try to reorganize the triple summation into the form of the LHS of (4))

(b) Based on the above lemma and the REINFORCE policy gradient (i.e., (P2)), show the PG expression (P3):

$$\nabla_\theta V^{\pi_\theta}(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[ Q^{\pi_\theta}(s, a) \nabla_\theta \log \pi_\theta(a|s) \right]. \quad (5)$$

(c) Let us check if the PG expressions used in some well-known papers are indeed correct.

- ACKTR, published in NeurIPS 2017 (available at <https://arxiv.org/abs/1708.05144>), is one variant

(a) *dep: Targeting expectation = State distribution expectation.*

$$\begin{aligned} \text{RHS: } & \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [\dot{V}(s, a)] \\ &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \left[ \sum_a \pi(a|s) \cdot \dot{V}(s, a) \right] \\ &= \frac{1}{1-\gamma} \sum_s d_\mu^{\pi_\theta}(s) \cdot \left[ \sum_a \pi(a|s) \cdot \dot{V}(s, a) \right], \text{ apply the definition of } d_\mu^{\pi_\theta} \\ &= \frac{1}{1-\gamma} \sum_s (1-\gamma) \cdot \left[ \sum_{\tau=0}^{\infty} \gamma^t \mathbb{P}(S_t = s | S_0, \pi) \cdot \left[ \sum_a \pi(a|s) \cdot \dot{V}(s, a) \right] \right] \\ &= \sum_s \frac{\sum_{\tau=0}^{\infty} \gamma^t \mathbb{P}(S_t = s | S_0, \pi)}{1-\gamma} \cdot \left[ \sum_a \pi(a|s) \cdot \dot{V}(s, a) \right] \\ &= \sum_s \sum_{\tau=0}^{\infty} \gamma^t \mathbb{P}(S_t = s | \pi) \cdot \left[ \sum_a \pi(a|s) \cdot \dot{V}(s, a) \right] \\ &= \sum_{\tau=0}^{\infty} \gamma^t \cdot \left( \sum_{s \in S} \sum_a \mathbb{P}(S_t = s, a | \pi) \cdot \dot{V}(s, a) \right) \\ &= \sum_{\tau=0}^{\infty} \gamma^t \mathbb{E}_{S_t, a_t \sim \pi_\theta} [\dot{V}(s, a)] = \mathbb{E}_{S_t, a_t \sim \pi_\theta} \left[ \sum_{\tau=0}^{\infty} \gamma^t \dot{V}(s, a) \right] \end{aligned}$$

$\stackrel{\text{dep}}{=} \mathbb{E}_{\tau \sim (S_0, a_0, s_1, a_1, \dots)}$

(b).

$$\text{Reinforce Policy gradient (P): } \nabla_\theta V^{\pi_\theta}(\mu) = \mathbb{E}_{\tau \sim \mu^{\pi_\theta}} \left[ \sum_{t=0}^{\infty} \gamma^t G_{t, \frac{1}{2} \Delta} \cdot \dot{V}_\theta(s_t) \right]$$

$\Rightarrow$  Lemma:  $\mathbb{E}_{\tau \sim \mu^{\pi_\theta}} \left[ \sum_{t=0}^{\infty} \gamma^t \dot{V}(s_t, a_t) \right] = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \left[ \sum_a \pi(a|s) \cdot \dot{V}(s, a) \right]$ .  $\dot{V}(s_t, a_t) \cdot G_{t, \frac{1}{2} \Delta} \cdot \dot{V}_\theta(s_t) \cdot \dot{V}_\theta(a_t) \cdot \dot{V}_\theta(s_t)$

Substitute in Lemma. We can get  $\nabla_\theta V^{\pi_\theta} = \frac{1}{1-\gamma} \mathbb{E}_{\tau \sim \mu^{\pi_\theta}} \left[ \sum_{t=0}^{\infty} \gamma^t \dot{V}_\theta(s_t) \cdot \dot{V}_\theta(a_t) \cdot \dot{V}_\theta(s_t) \right] \neq \mathbb{E}_{\theta} \mathbb{E}_{\mathcal{P}}$ .

(c).

- ACKTR,  $\nabla_\theta J(\theta) = \mathbb{E}_{\pi} \left[ \sum_{t=0}^{\infty} \gamma^t \dot{V}_\theta(s_t) \cdot \dot{V}_\theta(a_t) \cdot \dot{V}_\theta(s_t) \right]$ ,  $\dot{V}_\theta = G(s_t, a_t)$ .
- ACKR,  $\nabla_\theta J(\theta) = \mathbb{E}_{\pi} \left[ \sum_{t=0}^{\infty} \gamma^t \dot{V}_\theta(s_t) \cdot \dot{V}_\theta(a_t) \cdot \dot{V}_\theta(s_t) \right]$ ,  $\dot{V}_\theta = G(s_t, a_t)$ .

Wrong, look at discount factor.

$\Rightarrow \nabla_\theta J(\theta) = \mathbb{E}_{\pi} \left[ \sum_{t=0}^{\infty} \gamma^t \dot{V}_\theta(s_t) \cdot \dot{V}_\theta(a_t) \cdot \dot{V}_\theta(s_t) \right]$

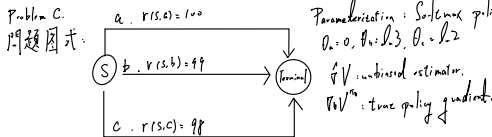
Consider an example similar to that in the slides of Lecture 6 for explaining the baseline. Suppose that there are only 1 non-terminal starting state (denoted by  $s$ ) and 3 actions (denoted by  $a, b, c$ ) in the MDP of interest. After applying any of the actions in the starting state  $s$ , the MDP would evolve from  $s$  to the terminal state, with probability 1. Moreover, consider the following setting:

- The rewards are deterministic and the reward function is defined as  $r(s, a) = 100$ ,  $r(s, b) = 99$ , and  $r(s, c) = 98$ . Moreover, there is no terminal reward.
- Let us consider a softmax policy with parameters  $\theta_a, \theta_b, \theta_c$  such that  $\pi_a(s) = \exp(\theta_a) / (\exp(\theta_a) + \exp(\theta_b) + \exp(\theta_c))$ . Moreover, the parameters are currently  $\theta_a = 0$ ,  $\theta_b = \ln 3$ , and  $\theta_c = \ln 2$ .
- We would like to combine PG with SGD. At each policy update, we would construct an unbiased estimate  $\hat{\nabla} V$  of the true policy gradient  $\nabla_{\theta} V^{\pi}$  by sampling one trajectory (Note:  $\nabla V$  is a random vector. In this example, each trajectory has only one time step, and  $s_0 = s$ ,  $s_1$  is either  $a, b$ , or  $c$ , and  $s_2$  is the terminal state).

(a) What are the mean vector of  $\hat{\nabla} V$  (denoted by  $E[\hat{\nabla} V]$ ) and the covariance matrix of  $\hat{\nabla} V$  (i.e.,  $E[(\hat{\nabla} V - E[\hat{\nabla} V])(\hat{\nabla} V - E[\hat{\nabla} V])^T]$ )?

(b) Suppose we leverage the value function  $V^{\pi}(s)$  as the baseline and denote by  $\hat{\nabla} V$  the corresponding estimated policy gradient. Then, what are the mean vector and the covariance matrix of  $\hat{\nabla} V$ ? (Note:  $\nabla V$  is also a random vector)

(c) Let  $B(s)$  denote a baseline policy gradient and  $\hat{\nabla} V_B$  be the corresponding estimated policy gradient ( $\nabla V_B$  is again a random vector). Suppose we say that a baseline function  $B(s)$  is optimal if it attains the minimum trace of the corresponding covariance matrix of  $\hat{\nabla} V_B$  among all possible state-dependent baselines. Please try to find one such optimal  $B(s)$ .



1. (a.)  
 $\pi_a(s) = \frac{\exp(\theta_a)}{\exp(\theta_a) + \exp(\theta_b) + \exp(\theta_c)} = \frac{e^0}{e^0 + e^{\ln 3} + e^{\ln 2}} = \frac{1}{6}$   
 $\pi_b(s) = \frac{e^{\ln 3}}{e^0 + e^{\ln 3} + e^{\ln 2}} = \frac{1}{2}$   
 $\pi_c(s) = \frac{e^{\ln 2}}{e^0 + e^{\ln 3} + e^{\ln 2}} = \frac{1}{3}$

2.  $C$ : compute the score function for 3 different actions.

For action  $a$ :  
 $\frac{\partial \pi_a(s, a)}{\partial \theta(s, a)} = \frac{1}{\pi_a(s, a)} \cdot \frac{\partial \pi_a(s, a)}{\partial \theta(s, a)} = \frac{\exp(\theta_a) \cdot \exp(\theta_a)}{(\exp(\theta_a) + \exp(\theta_b) + \exp(\theta_c))^2} = \frac{\exp(2\theta_a)}{(\exp(\theta_a) + \exp(\theta_b) + \exp(\theta_c))^2}$   
 $\frac{\partial \pi_a(s, a)}{\partial \theta(s, a)} = \frac{1}{\pi_a(s, a)} \cdot \frac{\partial \pi_a(s, a)}{\partial \theta(s, a)} = \frac{1}{\frac{1}{6}} \cdot \frac{1}{6} = 1$   
 $\frac{\partial \pi_b(s, a)}{\partial \theta(s, a)} = \frac{1}{\pi_b(s, a)} \cdot \frac{\partial \pi_b(s, a)}{\partial \theta(s, a)} = \frac{1}{\frac{1}{2}} \cdot \frac{1}{2} = 1$   
 $\frac{\partial \pi_c(s, a)}{\partial \theta(s, a)} = \frac{1}{\pi_c(s, a)} \cdot \frac{\partial \pi_c(s, a)}{\partial \theta(s, a)} = \frac{1}{\frac{1}{3}} \cdot \frac{1}{3} = 1$

For action  $b$ :  
 $\frac{\partial \pi_b(s, b)}{\partial \theta(s, b)} = \frac{1}{\pi_b(s, b)} \cdot \frac{\partial \pi_b(s, b)}{\partial \theta(s, b)} = \frac{1}{\frac{1}{2}} \cdot \frac{1}{2} = 1$   
 $\frac{\partial \pi_a(s, b)}{\partial \theta(s, b)} = \frac{1}{\pi_a(s, b)} \cdot \frac{\partial \pi_a(s, b)}{\partial \theta(s, b)} = \frac{1}{\frac{1}{6}} \cdot \frac{1}{6} = 1$   
 $\frac{\partial \pi_c(s, b)}{\partial \theta(s, b)} = \frac{1}{\pi_c(s, b)} \cdot \frac{\partial \pi_c(s, b)}{\partial \theta(s, b)} = \frac{1}{\frac{1}{3}} \cdot \frac{1}{3} = 1$

Same for action  $c$ :  
 $\frac{\partial \pi_c(s, c)}{\partial \theta(s, c)} = \frac{1}{\pi_c(s, c)} \cdot \frac{\partial \pi_c(s, c)}{\partial \theta(s, c)} = \frac{1}{\frac{1}{3}} \cdot \frac{1}{3} = 1$   
 $\frac{\partial \pi_a(s, c)}{\partial \theta(s, c)} = \frac{1}{\pi_a(s, c)} \cdot \frac{\partial \pi_a(s, c)}{\partial \theta(s, c)} = \frac{1}{\frac{1}{6}} \cdot \frac{1}{6} = 1$   
 $\frac{\partial \pi_b(s, c)}{\partial \theta(s, c)} = \frac{1}{\pi_b(s, c)} \cdot \frac{\partial \pi_b(s, c)}{\partial \theta(s, c)} = \frac{1}{\frac{1}{2}} \cdot \frac{1}{2} = 1$

3. For the exact policy gradient:  
 $\nabla_{\theta} V^{\pi}(s) = E_{\pi} \left[ \sum_{t=0}^{\infty} \gamma^t \nabla_{\theta} r(s_t, a_t) \right]$

But  $\hat{\nabla} V$  is a random vector that we get by sampling & projecting.  
 In this case,  $\hat{\nabla} V = \hat{V}(s, a) \cdot \nabla_{\theta} \pi_a(s, a) + \hat{V}(s, b) \cdot \nabla_{\theta} \pi_b(s, b) + \hat{V}(s, c) \cdot \nabla_{\theta} \pi_c(s, c)$   
 so,  $\hat{V}(s, a) = 100$ , and no discount factor.

$\hat{\nabla} V = \begin{bmatrix} 100 \cdot \left[ \frac{1}{6} \right] + \hat{V}(s, b) \cdot \left[ \frac{1}{2} \right] + \hat{V}(s, c) \cdot \left[ \frac{1}{3} \right] \\ \hat{V}(s, a) \cdot \left[ \frac{1}{6} \right] + \hat{V}(s, b) \cdot \left[ \frac{1}{2} \right] + \hat{V}(s, c) \cdot \left[ \frac{1}{3} \right] \\ \hat{V}(s, a) \cdot \left[ \frac{1}{6} \right] + \hat{V}(s, b) \cdot \left[ \frac{1}{2} \right] + \hat{V}(s, c) \cdot \left[ \frac{1}{3} \right] \end{bmatrix}$   
 $E[\hat{\nabla} V] = E \left[ \sum_{i \in \{a, b, c\}} (V_i(s) - V(s)) \cdot (V_i(s) - V(s)) \right] = \begin{bmatrix} \frac{1}{6} \\ \frac{1}{2} \\ \frac{1}{3} \end{bmatrix}$   
 $\frac{1}{6} \times 100 \begin{bmatrix} \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \end{bmatrix} + \frac{1}{2} \times 99 \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} + \frac{1}{3} \times 98 \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$

Covariance Matrix:  $E[\hat{\nabla} V \hat{\nabla} V^T] - E[\hat{\nabla} V] E[\hat{\nabla} V]^T$   
 $= E[\hat{\nabla} V \hat{\nabla} V^T] - E[\hat{\nabla} V] E[\hat{\nabla} V]^T$   
 $E[\hat{\nabla} V \hat{\nabla} V^T] = \frac{1}{6} \begin{bmatrix} 100 \\ 100 \\ 100 \end{bmatrix} \begin{bmatrix} 100 \\ 100 \\ 100 \end{bmatrix}^T + \frac{1}{2} \begin{bmatrix} 99 \\ 99 \\ 99 \end{bmatrix} \begin{bmatrix} 99 \\ 99 \\ 99 \end{bmatrix}^T + \frac{1}{3} \begin{bmatrix} 98 \\ 98 \\ 98 \end{bmatrix} \begin{bmatrix} 98 \\ 98 \\ 98 \end{bmatrix}^T$   
 $= \begin{bmatrix} \frac{10000}{6} + \frac{9801}{2} + \frac{9604}{3} & \frac{9900}{6} + \frac{9801}{2} + \frac{9518}{3} & \frac{9800}{6} + \frac{9702}{2} + \frac{9402}{3} \\ \frac{9900}{6} + \frac{9801}{2} + \frac{9518}{3} & \frac{9801}{2} + \frac{9604}{3} + \frac{9402}{3} & \frac{9702}{2} + \frac{9518}{3} + \frac{9201}{3} \\ \frac{9800}{6} + \frac{9702}{2} + \frac{9402}{3} & \frac{9702}{2} + \frac{9518}{3} + \frac{9201}{3} & \frac{9604}{3} + \frac{9402}{3} + \frac{9000}{3} \end{bmatrix}$   
 $= \begin{bmatrix} 3317 & 1937 & 1415 \\ 1937 & 8 & 415 \\ 1415 & 415 & 2 \end{bmatrix}$   
 $E[\hat{\nabla} V] E[\hat{\nabla} V]^T = \begin{bmatrix} \frac{1}{6} \\ \frac{1}{2} \\ \frac{1}{3} \end{bmatrix} \begin{bmatrix} \frac{1}{6} & \frac{1}{2} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{36} & \frac{1}{12} & \frac{1}{18} \\ \frac{1}{12} & \frac{1}{4} & \frac{1}{6} \\ \frac{1}{18} & \frac{1}{6} & \frac{1}{9} \end{bmatrix}$   
 $\Rightarrow \text{Covariance Matrix} = \begin{bmatrix} 3317 & 1937 & 1415 \\ 1937 & 8 & 415 \\ 1415 & 415 & 2 \end{bmatrix} - \begin{bmatrix} \frac{1}{36} & \frac{1}{12} & \frac{1}{18} \\ \frac{1}{12} & \frac{1}{4} & \frac{1}{6} \\ \frac{1}{18} & \frac{1}{6} & \frac{1}{9} \end{bmatrix} = \begin{bmatrix} 3317 & 1937 & 1415 \\ 1937 & 8 & 415 \\ 1415 & 415 & 2 \end{bmatrix}$

(b).  
 $\hat{V}(s) = \sum_{a \in A} \pi_a(s) Q(s, a)$   
 $\hat{V}(s) = \frac{1}{6} \times 100 + \frac{1}{2} \times 99 + \frac{1}{3} \times 98$   
 $\hat{V}(s) = \frac{393}{6}$   
 $\Rightarrow \text{Now } R(s, a) = V(s, a) - \hat{V}(s)$   
 $R(s, b) = 99 - \frac{393}{6}$   
 $R(s, c) = 98 - \frac{393}{6}$   
 $E[\hat{\nabla} V] = \frac{1}{6} \cdot \frac{7}{6} \cdot \begin{bmatrix} \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \end{bmatrix} + \frac{1}{2} \cdot \frac{1}{6} \cdot \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} + \frac{1}{3} \cdot \frac{1}{6} \cdot \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$   
 $= \begin{bmatrix} \frac{7}{36} \\ \frac{1}{12} \\ \frac{1}{18} \end{bmatrix}$   
 $\Rightarrow \text{Mean stay the same.}$

Variance:  $E[\hat{\nabla} V \hat{\nabla} V^T] - E[\hat{\nabla} V] E[\hat{\nabla} V]^T$   
 $= E[\hat{\nabla} V \hat{\nabla} V^T] - E[\hat{\nabla} V] E[\hat{\nabla} V]^T$   
 $E[\hat{\nabla} V \hat{\nabla} V^T] = \frac{1}{6} \begin{bmatrix} 7 \\ 7 \\ 7 \end{bmatrix} \begin{bmatrix} 7 \\ 7 \\ 7 \end{bmatrix}^T + \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}^T + \frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}^T$   
 $= \begin{bmatrix} \frac{49}{6} + \frac{1}{2} + \frac{1}{3} & \frac{7}{12} + \frac{1}{4} + \frac{1}{6} & \frac{7}{18} + \frac{1}{6} + \frac{1}{9} \\ \frac{7}{12} + \frac{1}{4} + \frac{1}{6} & \frac{1}{2} + \frac{1}{4} + \frac{1}{6} & \frac{1}{6} + \frac{1}{4} + \frac{1}{6} \\ \frac{7}{18} + \frac{1}{6} + \frac{1}{9} & \frac{1}{6} + \frac{1}{4} + \frac{1}{6} & \frac{1}{3} + \frac{1}{4} + \frac{1}{6} \end{bmatrix}$   
 $= \begin{bmatrix} \frac{71}{6} & \frac{17}{12} & \frac{7}{9} \\ \frac{17}{12} & \frac{7}{6} & \frac{7}{9} \\ \frac{7}{9} & \frac{7}{9} & \frac{7}{6} \end{bmatrix}$

$$\text{Variance} = \mathbb{E}[\tilde{V}V \cdot \tilde{V}V^T] - \mathbb{E}[\tilde{V}V] \cdot \mathbb{E}[\tilde{V}V]^T$$

$$= \begin{bmatrix} \frac{71}{432} \\ \frac{7}{144} \\ \frac{7}{54} \end{bmatrix} - \begin{bmatrix} \left(\frac{7}{36}\right)^2 \\ \left(\frac{1}{12}\right)^2 \\ \left(\frac{5}{18}\right)^2 \end{bmatrix} = \begin{bmatrix} \frac{41}{324} \\ \frac{1}{9} \\ \frac{17}{324} \end{bmatrix} \approx \begin{bmatrix} 0.1265 \\ 0.1111 \\ 0.0525 \end{bmatrix} \quad \text{H}$$

(C)

1° Minimize the trace of corresponding

covariance matrix of  $\tilde{V}V$ .

$$\text{Tr}(C \cdot V(\tilde{V}V)) = \mathbb{E}_{a \sim \pi} [\|(\beta_{1:a}, a) - b\| \cdot \mathbb{E}_0 \mathbb{E}_{\pi} [\|a(s)\|]^2] \cdot \mathbb{E}[\tilde{V}V]$$

Since  $\mathbb{E}[\tilde{V}V]$  have nothing to do with base line.

$$\text{our goal is to minimize } \mathbb{E}_{a \sim \pi} [\|(\beta_{1:a}, a) - b\| \cdot \mathbb{E}_0 \mathbb{E}_{\pi} [\|a(s)\|]^2] = J(b)$$

$$2^\circ \frac{dJ(b)}{db} = \mathbb{E}_{a \sim \pi} [2 \cdot (\beta_{1:a}, a) - b] \cdot (1) \cdot \mathbb{E}_0 \mathbb{E}_{\pi} [\|a(s)\|]^2 = 0$$

Since expectation value is a linear operator, we can get

$$\Rightarrow \mathbb{E}_{a \sim \pi} [\beta_{1:a}, a] \cdot \mathbb{E}_0 \mathbb{E}_{\pi} [\|a(s)\|]^2 - \mathbb{E}_{a \sim \pi} [b \cdot \mathbb{E}_0 \mathbb{E}_{\pi} [\|a(s)\|]^2] = 0. \quad \text{to have nothing to do with action, so.}$$

$$\Rightarrow \mathbb{E}_{a \sim \pi} [\beta_{1:a}, a] \cdot \mathbb{E}_0 \mathbb{E}_{\pi} [\|a(s)\|]^2 - b \cdot \mathbb{E}_{a \sim \pi} [\mathbb{E}_0 \mathbb{E}_{\pi} [\|a(s)\|]^2] = 0$$

$$\Rightarrow b^* = b = \frac{\mathbb{E}_{a \sim \pi} [\beta_{1:a}, a] \cdot \mathbb{E}_0 \mathbb{E}_{\pi} [\|a(s)\|]^2}{\mathbb{E}_{a \sim \pi} [\mathbb{E}_0 \mathbb{E}_{\pi} [\|a(s)\|]^2]}$$

though this result, we can observe that the optimal baseline isn't  $\mathbb{E}[\beta_{1:a}, a]$ . Result shows that for every action's  $a$  have it's own weight  $\mathbb{E}_0 \mathbb{E}_{\pi} [\|a(s)\|]^2$ , which means the sensitivity of the action for the model parameters, if one action  $a$  has a big gradient vector, which means when sampling to that action, the change will be more massively. To avoid the sudden change, the baseline is hard to align with these actions.